Error Semitransparent Universal Control of a Bosonic Logical Qubit

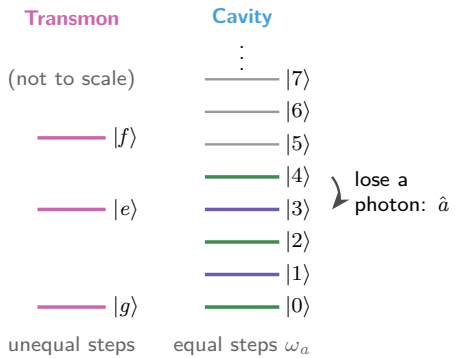
Error transparency and pulse optimisation

Ziyi Dong

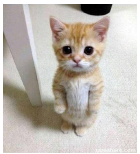
Presenting Roy, Wetherbee & Fatemi, arXiv:2603.15356
with our simulation replication

November 6, 2026

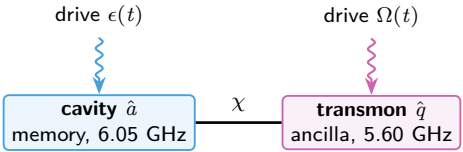
What is a bosonic qubit?

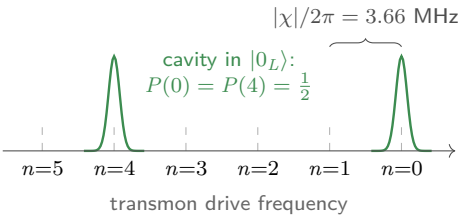


- **A qubit is two states.** A transmon uses its lowest two, g and e.
- **A bosonic qubit** stores them in a harmonic oscillator (a 3D cavity), spread over several photon-number (Fock) states $|n\rangle$.
- **Why?** The cavity lives longer ($T_1 = 180$ vs 70 μs here), and its main error is one kind: losing a photon, $\hat{a}$. The extra levels give room to *detect* it and undo it.
- **This talk: the binomial "kitten" code** (green rungs), $|0_L\rangle = (|0\rangle + |4\rangle)/\sqrt{2}$, $|1_L\rangle = |2\rangle$.



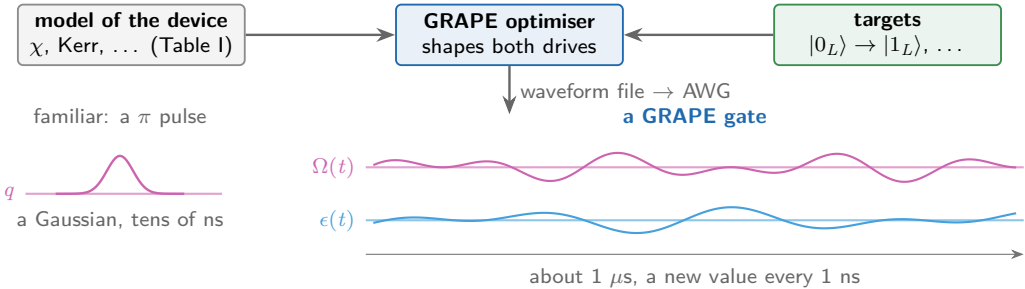
How we steer a cavity: the transmon and χ





- **The transmon line splits by photon number:** it sits at $\omega_q + n\chi$.
- A transmon pulse longer than $\sim 1/|\chi| \approx 270$ ns hits **one photon number only**.
- **Only two drives, both linear:** $\epsilon(t)$ pushes the cavity, $\Omega(t)$ rotates the transmon. All the nonlinearity comes from χ.

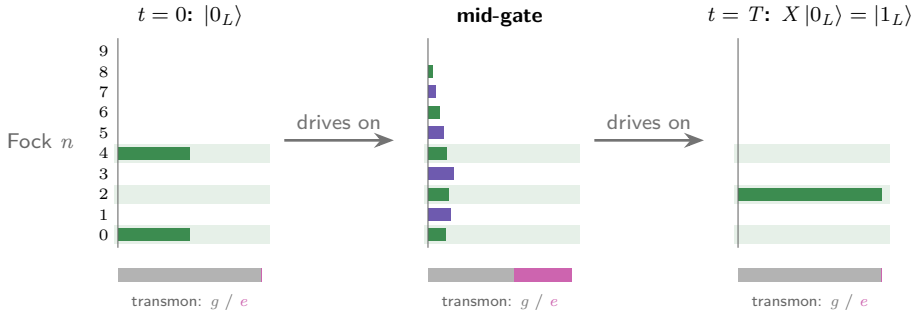
How the gates are made: optimal control (GRAPE)



- **Same hardware as a π pulse:** only the waveform changes, chosen by a computer, every ns, on both drives.
- **Model based:** designed on the device model, played as is. Wrong χ or Kerr, wrong gate.
- **Asked only about the endpoints:** what the state does in between is left free.

The loop in detail: slide 32.

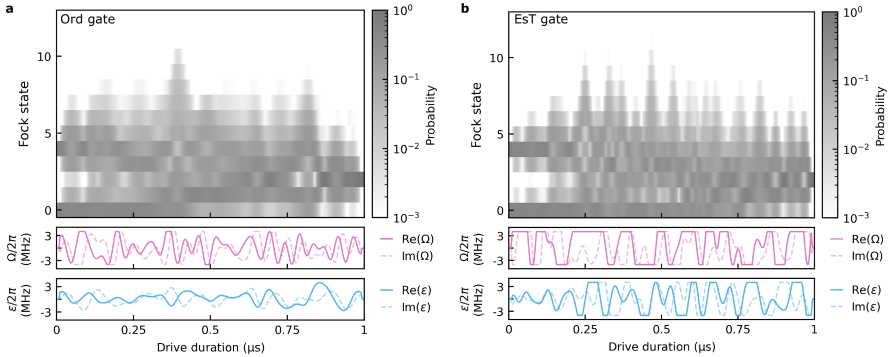
What happens to the code space during a gate



- **Before and after:** a codeword. Only Fock 0, 2, 4 (green stripes), even parity, transmon in g.
- **During:** the drives spread the state over many Fock states, odd and even, and entangle it with the transmon. It is *not* a codeword, and its parity is not defined.
- **Why:** the cavity drive is the only thing that moves the cavity states, and the gate needs χ to treat each photon number differently. So it has to spread the state out first.

Cartoon with made-up numbers. Real data on the next slide.

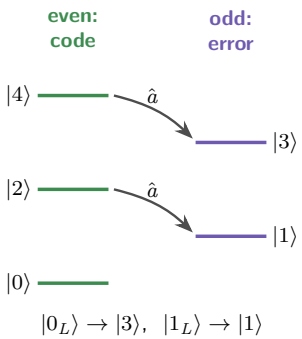
The same thing in the paper's simulation

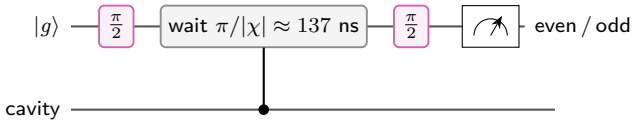


Photon-number occupation (grey, log scale) during the paper's 1 μs X gate, with the transmon drive Ω and cavity drive ϵ below. (a) an ordinary (Ord) gate, (b) the error-semitransparent (EsT) gate. **Both reach Fock $\sim$10 mid-gate**, although the code uses only Fock 0–4.

Roy et al., Fig. A5.

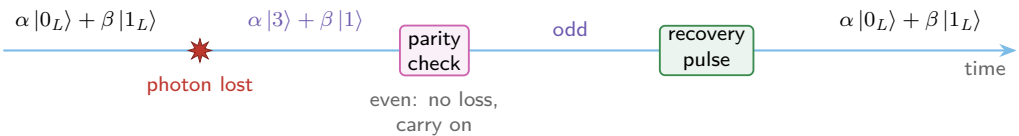
Correcting one photon loss at rest (1/2): parity





- One photon lost flips the parity: even $\to$ odd.
- During the wait the transmon picks up a phase $n\pi$. Even and odd n leave it in **opposite** states, so one readout gives the parity.
- It learns the parity, **not** n and not the logical state.

Correcting one photon loss at rest (2/2): the cycle

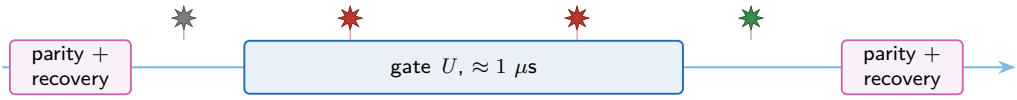


Why it works

$\hat{a}\,|0_L\rangle = \sqrt{2}\,|3\rangle$ and $\hat{a}\,|1_L\rangle = \sqrt{2}\,|1\rangle$: the **same** $\sqrt{2}$, because both codewords hold $\bar{n} = 2$ photons on average. So α and β come through unchanged. The loss tells the environment *that* a photon left, not *which* codeword it left from.

The paper's recovery: a 1 μs optimised pulse that dumps the error into the transmon, then a reset.

The problem: a photon lost *during* a gate



before: the gate then acts on an error word — fine only if it works there too

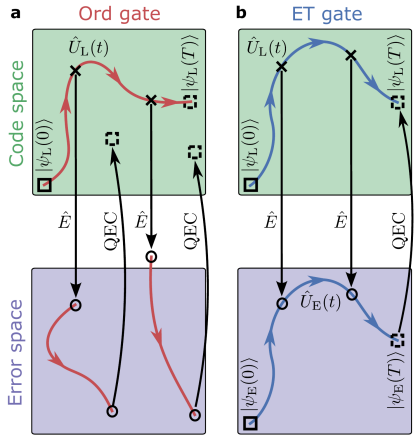
during: lands somewhere different for every loss time t: **not correctable**

after: parity catches it, recovery fixes it, as at rest

- Photon lost at t: the rest of the gate, $U(T,t)$, moves it to a t-dependent place. t is random, and the recovery has **one** fixed target.
- **How often?** About $\bar{n}\,T/T_1 = 2\times 1\ \mu\text{s}/180\ \mu\text{s} \approx 1\%$ of gates lose a photon mid-gate.

In plain words: a photon lost while idling is correctable; a photon lost during an ordinary gate, in general, is not.

What the paper aims for: error-transparent gates

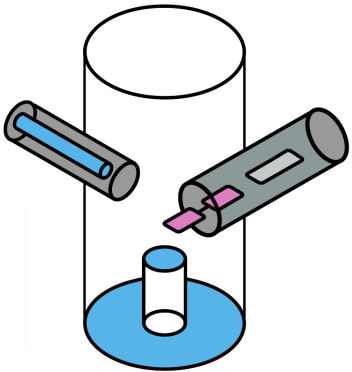


Roy et al., Fig. 1a–b.

- **(a) Ordinary gate.** A loss ($\times$) at each time lands at a different point in the error space; QEC brings each back to a different, wrong code state.
- **(b) Error-transparent gate.** The error space is carried along the same path as the code space, so every loss ends at the same point and one QEC step fixes it.
- **This paper:** X, H, T with ordinary linear drives, only *semi*-transparent. About $5\times$ lower infidelity when a photon is lost mid-gate (measured).

Break-even: Ni et al. and Sivak et al., Nature 616 (2023).

The device: a 3D cavity package



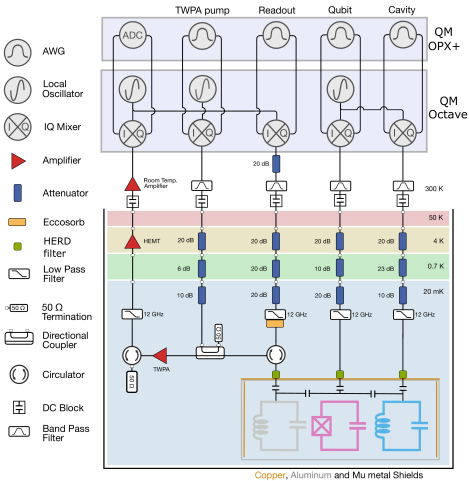
Roy et al., Fig. A1.

- **Cavity** (light blue): a $\lambda/4$ coaxial cavity machined from 4N (99.99%) aluminium. Holds the logical qubit. 6.05 GHz, $T_1 = 180$ μs.
- **Transmon** (pink): Nb pads and an Al/AlO_x/Al junction on a silicon chip, inserted through a side port. The ancilla. 5.60 GHz, $T_1 = 70$ μs.
- **Readout resonator** (grey, same chip): 8.92 GHz, reads the transmon.
- **Shielding:** OFHC copper bracket, black-coated copper shim, indium-sealed Al can, mixing-chamber can.

Etched cavities like this usually reach near 1 ms; this one degraded in shipping and storage.

Roy et al., App. B (same package as their Ref. [25]). Coaxial cavities: Reagor et al., PRB 94, 014506 (2016).

The wiring



Roy et al., Fig. A2 and App. B.

1. **Pulses** (top): a Quantum Machines OPX+ makes the I/Q waveforms; the Octave mixes them up to GHz with local oscillators. Transmon and cavity drives share one LO.
2. **Input lines** (qubit, cavity, readout): 50–60 dB of cold attenuation at 4 K, 0.7 K and 20 mK, then 12 GHz low-pass and HERD filters (Eccosorb on the readout line), to keep thermal photons out.
3. **Readout chain**: circulator → pumped TWPA at 20 mK → HEMT at 4 K → room-temperature amplifier → ADC.
4. **Shields**: copper, aluminium, mu-metal.

Device parameters

Quantity		Value	Hamiltonian term
Cavity frequency	$\omega_a/2\pi$	6.05 GHz	$\omega_a\,\hat{a}^\dagger\hat{a}$
Readout frequency	$\omega_r/2\pi$	8.92 GHz	$\omega_r\,\hat{r}^\dagger\hat{r}$
Qubit frequency	$\omega_q/2\pi$	5.60 GHz	$\omega_q\,\hat{q}^\dagger\hat{q}$
Qubit–readout dispersive	$\chi_r/2\pi$	-600 kHz	$\chi_r\,\hat{r}^\dagger\hat{r}\,\lvert e\rangle\langle e\rvert$
Qubit–cavity dispersive	$\chi/2\pi$	-3.66 MHz	$\chi\,\hat{a}^\dagger\hat{a}\,\lvert e\rangle\langle e\rvert$
Qubit anharmonicity	$K_q/2\pi$	-180 MHz	$K_q\,\hat{q}^{\dagger 2}\hat{q}^2/2$
2nd-order dispersive	$\chi'/2\pi$	39 kHz	$\chi'\,\hat{a}^{\dagger 2}\hat{a}^2\,\lvert e\rangle\langle e\rvert/2$
Cavity self-Kerr	$K/2\pi$	-22 kHz	$K\,\hat{a}^{\dagger 2}\hat{a}^2/2$
2nd-order self-Kerr	$K'/2\pi$	590 Hz	$K'\,\hat{a}^{\dagger 3}\hat{a}^3/6$

Quantity		Value
Qubit lifetime	T_{1q}	70 μs
Qubit Ramsey	$T_{2q,R}$	30 μs
Qubit Hahn echo	$T_{2q,E}$	84 μs
Cavity lifetime	T_{1a}	180 μs
Cavity Ramsey	$T_{2a,R}$	290 μs
Qubit thermal pop.		1.5%
Cavity thermal pop.		0.9%

- $|\chi|/2\pi \approx 3.7$ MHz $\gg$ every linewidth (qubit $\sim$10 kHz, cavity $\sim$1 kHz): photon-number peaks are well resolved.
- Kerr terms are tiny (kHz) but not zero. They come back as the reason transparency is only "semi".

Roy et al., Table I. The qubit couples to a two-level system that splits its line by 50–60 kHz. Next: the same terms as one Hamiltonian.

The device: a cavity dispersively coupled to a transmon

$\hat{a}$: cavity (storage oscillator, holds the logical qubit). $\hat{q}$: transmon (ancilla, for control and parity readout). In the dispersive regime:

$$H(t) = H_a + H_q + H_{\text{int}} + H_{\text{drive}}(t) \qquad \text{[Eq. A1]}$$

$$H_a/\hbar = \omega_a\, \hat{a}^\dagger \hat{a} + \tfrac{K}{2}\, \hat{a}^{\dagger 2}\hat{a}^2 + \tfrac{K'}{6}\, \hat{a}^{\dagger 3}\hat{a}^3 \qquad \text{[A2]}$$

$$H_q/\hbar = \omega_q\, \hat{q}^\dagger \hat{q} + \tfrac{K_q}{2}\, \hat{q}^{\dagger 2}\hat{q}^2 \qquad \text{[A3]}$$

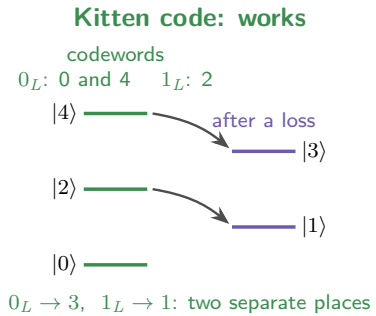
$$H_{\text{int}}/\hbar = \chi\, \hat{a}^\dagger \hat{a}\, \hat{q}^\dagger \hat{q} + \tfrac{\chi'}{2}\, \hat{q}^\dagger \hat{q}\, \hat{a}^{\dagger 2}\hat{a}^2 \qquad \text{[A4]}$$

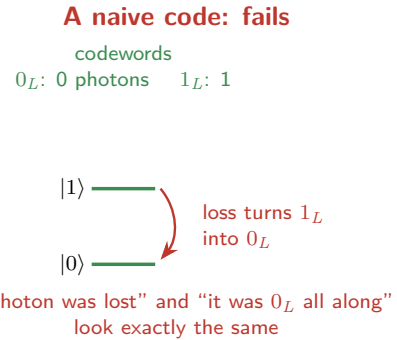
$$H_{\text{drive}}(t)/\hbar = \epsilon(t)\, \hat{a} + \Omega(t)\, \hat{q} + \text{h.c.} \qquad \text{[A5]}$$

$\chi/2\pi$	-3.66 MHz	$K/2\pi$	-22 kHz	$K_q/2\pi$	-180 MHz
$\chi'/2\pi$	39 kHz	$K'/2\pi$	590 Hz	cavity T_1	180 μs

Main-text Eq. 9 is the same Hamiltonian at low order: it drops K' and χ'. Values from Table I.

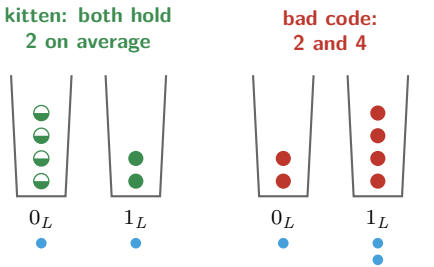
Rule 1: a loss must not mix up the codewords

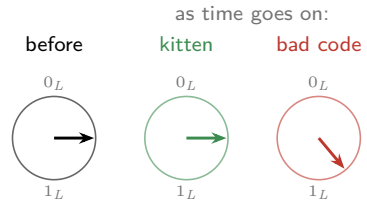




In plain words: after a loss, the two codewords must still be told apart, and the error must land somewhere the code itself does not use, so a parity check can see it.

Rule 2: a loss must not reveal which codeword it hit

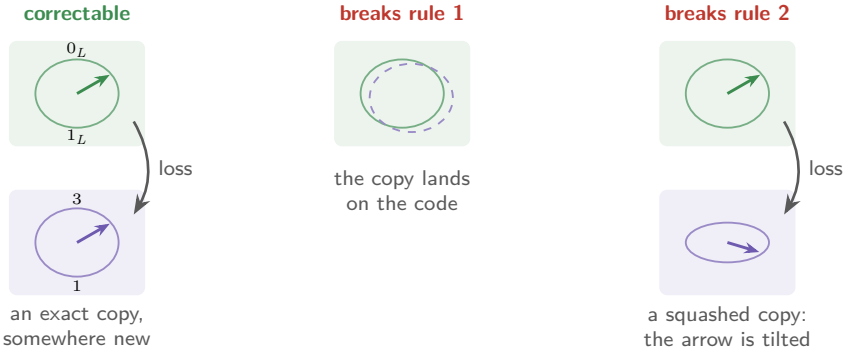




The logical state as an arrow. Equal leak rates leave it alone; unequal rates **tilt it** toward the leakier codeword.

In plain words: a lost photon is a clue. If one codeword leaks faster, the clue says "probably that one", and the qubit is partly measured. Both codewords must hold the same number of photons on average.

Knill–Laflamme is the two rules

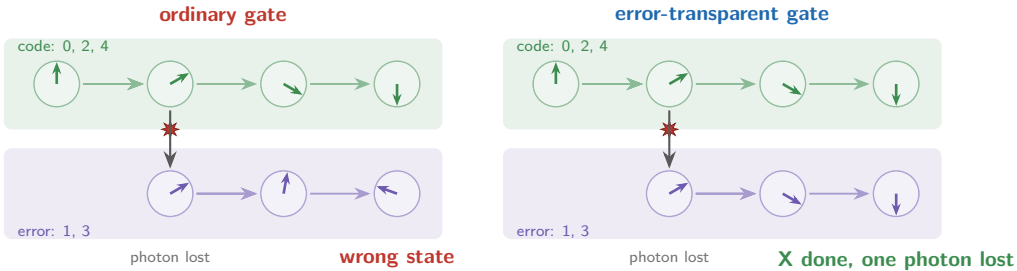


The Knill–Laflamme theorem

It defines the condition for error correction: $\left\langle i_L \middle| E_a^\dagger E_b \middle| j_L \right\rangle = \alpha_{ab}\,\delta_{ij}$.

δ_{ij} is rule 1; α_{ab} the same for both codewords is rule 2.

Error transparent: the gate runs the same on both tracks



- The arrow is the logical state. A loss only **moves** it to the error track, pointing the same way (the two rules).
- **Ordinary gate:** on the error track the drives turn it some other way, and where it ends depends on when the photon left.
- **Transparent gate:** the error track turns **in step** with the code track, so a loss at any time ends the same way. One recovery fixes them all.

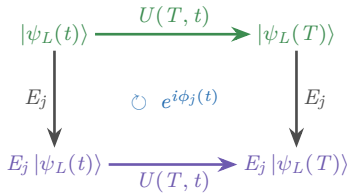
Error transparency: the jump commutes through the gate

The demand: for all $t \in [0, T]$,

$$U(T,t)\, E_j\, U(t,0)\, |\psi_L(0)\rangle = e^{i\phi_j(t)}\, E_j\, U(T,0)\, |\psi_L(0)\rangle \qquad \text{[Sec. II]}$$

- **Left:** evolve to t, get hit by E_j, then finish the gate.
- **Right:** finish the whole ideal gate, then get hit, times a scalar phase.

If it holds $\forall t$, a photon lost at *any* time gives the **ideal gate output with one error applied at the end**: a single, known, correctable configuration. One recovery at $t = T$ cleans it up.



Both routes reach the same state, up to $e^{i\phi_j(t)}$, for *every* t.

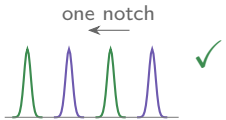
Which parts of the device let a loss pass through?

the cavity drive



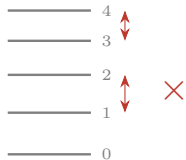
A push does not care how many photons are there. Push then lose = lose then push.

χ



Every photon shifts the transmon by the same χ. After a loss, everything is shifted by one notch: known, so correctable.

Kerr (and χ')



A photon's energy depends on how many others are there. Losing one from 4 shifts the state's frequency by a different amount than losing one from 2.

The same, term by term with formulas: slides B10–B11 (paper App. A).

Why Kerr spoils it: the logical phase drifts





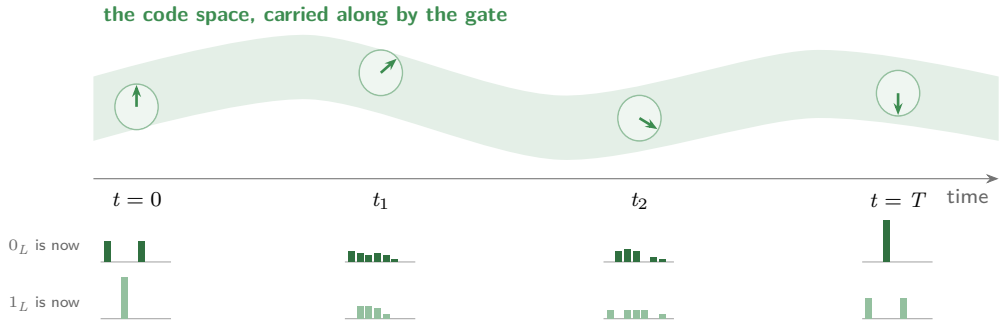
logical arrow after the gate:
early loss turned most,
late loss turned least

- After a loss the two error words differ in frequency by a **different amount** than the codewords did, so the arrow turns by an amount set by **how long the photon has been gone**.
- The loss time is random: the arrows **fan out**, part of the qubit is lost.

Why linear drives cannot cancel it

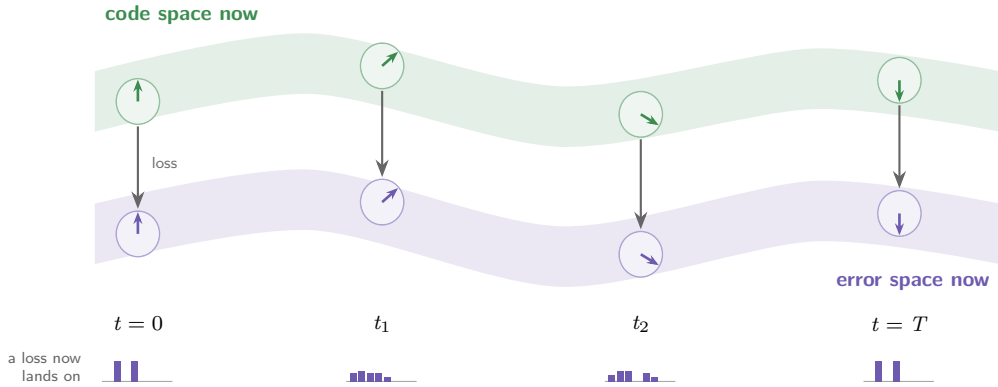
That needs something that grows with photon number the way Kerr does. The drives ignore photon number; χ is the same per photon. So GRAPE keeps the damage *small*, never zero: error **semi**transparent.

The instantaneous code space: where the codewords are right now



- The gate carries both codewords along. At each moment, the two states they have become **are** the code, right now.
- The qubit is still there, any arrow is carried along; it is just written in different photon numbers. At $t = T$ the code is back on Fock 0, 2, 4.

The instantaneous error space: where a loss right now would send you



- At each moment, take the code *as it is now* and remove one photon. Where that lands is the error space **now**. While idling it is just Fock 1 and 3.
- Transparent, in this picture: the gate carries the purple ribbon along **with** the green one, and turns the arrows on both the same way.

Three checks, one number each

1. Is a loss right now correctable?



QEC violation
0 = yes

2. Does a lost-photon state stay on its ribbon?



leakage
0 = stays

3. Do the two arrows agree?



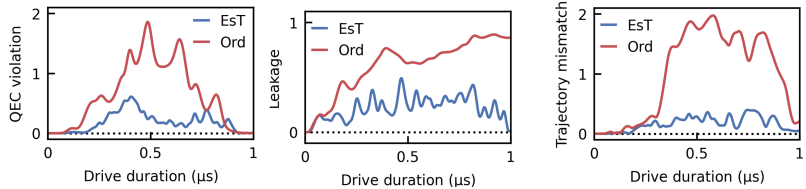
trajectory mismatch
0 = agree

- Each check gives a number at every moment of the gate; **all three zero, all the time** means fully transparent.
- Linear drives cannot make them all zero (Kerr). The optimiser makes them small. The paper plots these three curves for its X gate (Fig. 1d–f); so do our results slides.

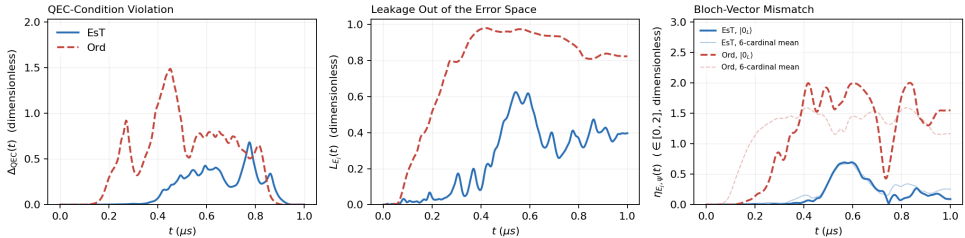
The three conditions and metrics as formulas (paper Eqs. 1–8): slides B12–B21.

Our X gate vs the paper: the three metrics in time

Paper, Fig. 1d–f



Ours, Fig. 2a



Left to right: Δ_{QEC}, L, η for $|0_L\rangle$ (ours also shows the six-cardinal mean, faint). X gate, $1\,\mu$s, lossless simulation, linear axes. EsT blue, Ord red.

What agrees, and what does not

X gate, EsT / Ord	Paper	Ours
Δ_{QEC}, time-averaged	0.18 / 0.65	0.156 / 0.481
L, time-averaged	0.22 / 0.61	0.258 / 0.740
$\eta(\lvert 0_L\rangle)$, time-averaged	0.17 / 0.99	0.169 / 1.11
$L(T)$, end of gate	0.013 / 0.862	**0.396** / 0.824
Δ_{QEC}, peak	$\approx$0.6 / $\approx$1.9	0.68 / 1.49
$\eta(\lvert 0_L\rangle)$ at $t = T$	$\approx$0.05 / $\approx$0.2	0.09 / **1.55**

Paper time averages and $L(T)$: digitised from Fig. 1d–f. $\approx$: read by eye.

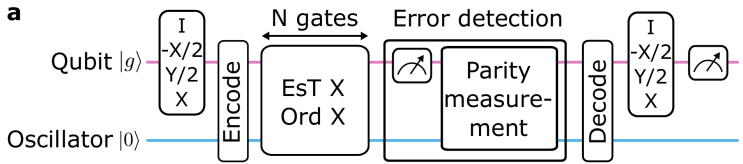
Agrees

- EsT below Ord through mid-pulse, on all three.
- All three time averages within about 25% of the paper's.

Differs

- The end of the gate: their EsT finishes transparent, ours still leaks.
- Our late Δ_{QEC} peak (0.78 μs).

How they test a gate in the lab

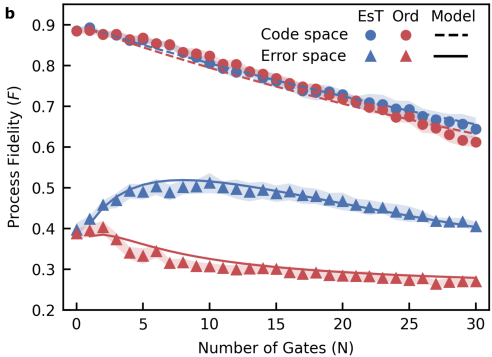


Roy et al., Fig. 3a.

1. **Prepare and encode.** A transmon rotation picks a test state; the encode pulse moves it into the cavity.
2. **Apply N gates** (EsT or Ord), back to back.
3. **Sort the shot.** Read the transmon and keep only shots that end in g. Then parity: even = **code space**, odd = **error space** (a photon was lost somewhere in the N gates).
4. **Decode and score.** Map the cavity back onto the transmon (one decode for each parity) and do process tomography, separately for the two sorts.

Error-space fidelity: in the shots that lost a photon, how much of the logical state survived. 0.25 means nothing did.

X gate: same in code space, better in error space



Roy et al., Fig. 3b (measured).

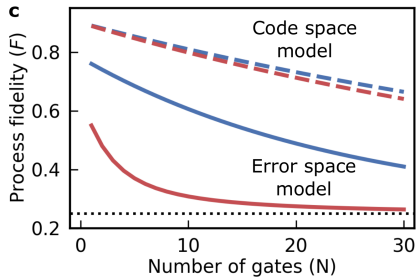
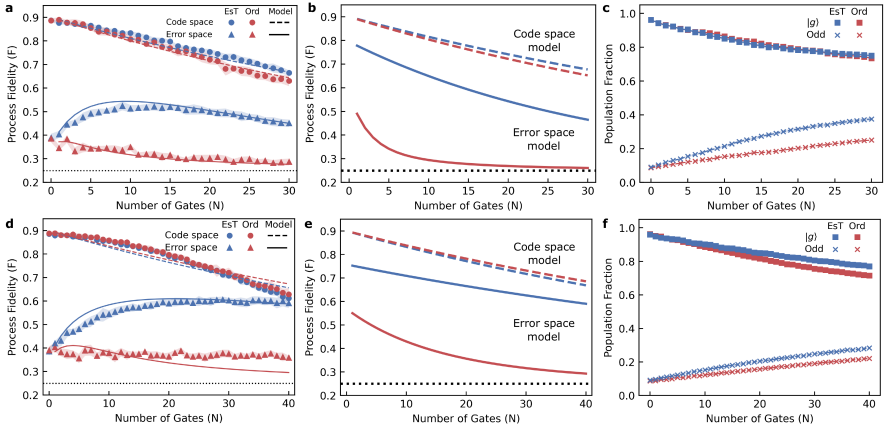


Fig. 3c (model, without parity-readout errors).

- **Code space (circles):** EsT and Ord the same; transparency costs nothing here.
- **Error space (triangles):** EsT ≈ 0.5, Ord ≈ 0.3 near 10 gates (0.25 = all lost). The early rise is parity-readout and state-prep error, removed in the model on the right.

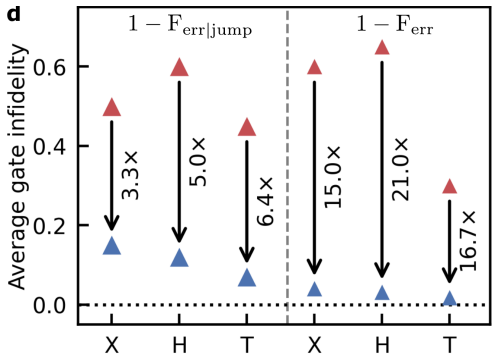
H and T gates: the same pattern



Roy et al., Fig. A7 (measured, with model). Top: H. Bottom: T.

Left: data, as for X. **Middle:** model without parity-readout errors. **Right:** fraction of shots with the transmon in g (squares) and with odd parity (crosses), which grows as losses pile up.

Per gate: about $5\times$ less infidelity after a loss

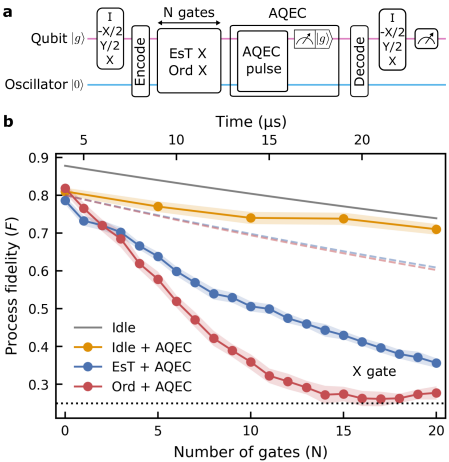


Roy et al., Fig. 3d. Red Ord, blue EsT.

- **Left half:** infidelity of one gate during which a photon is lost (it starts in the code, ends in the error space). EsT is $3.3\times$, $5.0\times$, $6.4\times$ lower for X, H, T: **about $5\times$ on average**.
- **Right half:** infidelity of one gate run entirely in the error space. 15–$21\times$ lower.

Per-gate numbers are extracted from the decay curves with the paper's model (its App. F).

With error correction, in the lab



Roy et al., Fig. 4a–b (measured, X gate).

- **Sequence:** N gates, then **one** correction step: a 1 μs AQEC pulse moves error states back to the code and dumps the error into the transmon, which is then reset (1.2 μs).
- **EsT + AQEC stays above Ord + AQEC**, by up to about 0.17 near 10 gates (their Fig. 4c).
- Both fall below idling + AQEC: two losses, transmon errors and control errors are not corrected, and they pile up with N.

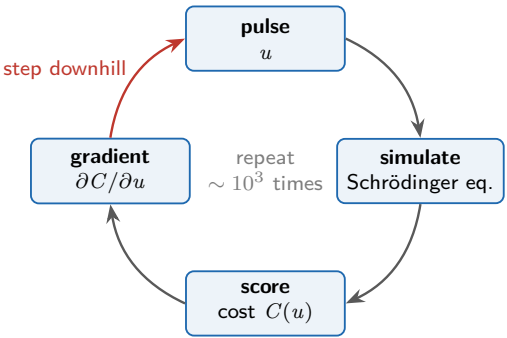
What GRAPE does (Gradient Ascent Pulse Engineering)

The pulse as a list of numbers.

Cut the pulse into N steps of $\Delta t = 1$ ns, constant on each step. Each step holds 4 real numbers: I and Q of the cavity and transmon drives (slide 34). Stack them end to end:

$$u = \big(u(t_1),\, u(t_2),\, \ldots,\, u(t_N)\big),$$

a single vector of $4N$ numbers. [App. C]



In plain words: GRAPE is guided trial and error. Simulate the pulse, score how far the test states land from their targets, and let the gradient say which way to nudge each of the $4N$ numbers. All are nudged together. It is a local search: it usually converges in about a thousand rounds, but where it lands depends on the random start.

The model the optimiser simulates

Go to the frame rotating with both modes, drives on resonance. Then ω_a, ω_q drop out, and what is left of slide 14 is the static **drift**:

$$H_0/\hbar = \chi\, \hat{a}^\dagger \hat{a}\, \hat{q}^\dagger \hat{q} + \tfrac{K}{2}\, \hat{a}^{\dagger 2}\hat{a}^2 + \tfrac{K'}{6}\, \hat{a}^{\dagger 3}\hat{a}^3 + \tfrac{\chi'}{2}\, \hat{q}^\dagger \hat{q}\, \hat{a}^{\dagger 2}\hat{a}^2 + \tfrac{K_q}{2}\, \hat{q}^{\dagger 2}\hat{q}^2.$$

[from Eqs. A2–A4]

cavity	Fock $\lvert 0\rangle, \ldots, \lvert 19\rangle$ $(n_c = 20)$	pulses reach $\sim \lvert 10\rangle$; stable for 16–28
transmon	$\lvert g\rangle, \lvert e\rangle, \lvert f\rangle$	keeps leakage to $\lvert f\rangle$
total	dimension 60	[our truncation]

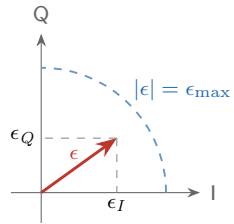
Lossless and unitary

Dissipation is not simulated: the evolution is purely unitary. Photon loss appears only inside the cost, as the operator $\hat{a}$.

Rewriting the drive as real controls: I and Q

Split each complex drive (Eq. 10) into two real parts, $\epsilon = \epsilon_I + i\,\epsilon_Q$ and $\Omega = \Omega_I + i\,\Omega_Q$:

$$\epsilon\,\hat{a} + \epsilon^*\,\hat{a}^\dagger = \epsilon_I\,(\hat{a} + \hat{a}^\dagger) + \epsilon_Q\; i(\hat{a} - \hat{a}^\dagger).$$



Same for the transmon. So

$$H(t) = H_0 + \sum_{j=1}^{4} u_j(t)\,H_j,$$

$$u = (\epsilon_I,\ \epsilon_Q,\ \Omega_I,\ \Omega_Q),$$
$$H_j = \big(\hat{a} + \hat{a}^\dagger,\ i(\hat{a} - \hat{a}^\dagger),\ \hat{q} + \hat{q}^\dagger,\ i(\hat{q} - \hat{q}^\dagger)\big).$$

In plain words: the optimiser only sees real numbers, and H is linear in them. I and Q are also exactly the two channels of the IQ mixer that makes the pulse in the lab.

Piecewise-constant pulses and the gradient

Hold u fixed on each step of length Δt. On step k:

$$H_k = H_0 + \sum_j u_j(k)\, H_j, \qquad U_k = e^{-iH_k \Delta t}, \qquad U(T,0) = U_N \cdots U_2\, U_1.$$



- The pulse is constant on each step, so each step is one matrix exponential.
- The whole gate is these N matrices multiplied in order.
- Each number in u affects only its own step, so one sweep forward and one sweep back through the product give the gradient for all $4N$ numbers at once.

What goes in, and what should come out

Inputs. The six cardinal states of the logical Bloch sphere, transmon in $|g\rangle$,

$$|\pm Z_L\rangle\,, \quad |\pm X_L\rangle\,, \quad |\pm Y_L\rangle\,,$$

and their error partners $|\psi_E\rangle \propto \hat{a}\,|\psi_C\rangle$ in the error space (ψ_C: a code state, the ψ_L of Part I).

Targets. The ideal logical gate applied to each input, with the transmon back in $|g\rangle$.

Gates and time grid

gate	length	drives	steps N
X	$1\ \mu$s	cavity + transmon	1000
H	$1\ \mu$s	cavity + transmon	1000
T	$0.6\ \mu$s	transmon only	600

Piecewise constant, $\Delta t = 1$ ns. [App. C]

In plain words: every gate is tested on six states at once. A pulse that sends all six to the right place has the right logical action, relative phases included.

The cost function: one weighted sum

$$C_{\text{tot}} = \sum_i w_i \, C_i \qquad \text{[Eq. C8]}$$

Term	Question it asks	Evaluated
C_1 gate infidelity	Did the gate happen?	end of pulse
C_2 transparency	Does a loss *now* land where an earlier one does?	every step
C_3 velocity variance	Is the motion spread over the gate?	every step
C_4 amplitude cap	Is the drive within 4 MHz?	every step [ours]

Eqs. 6–8 are diagnostics, not the objective

The optimiser uses **state overlaps**, which are easy to differentiate; the metrics only check the result. Not one cost per condition: C_2 pushes on Conditions 2 and 3 together, nothing targets Condition 1 directly, and C_3 is a regulariser.

C_1: did the gate happen?

$$C_1 = 1 - F_{\text{target}} = 1 - \frac{1}{6}\sum_{i=1}^{6} \left| \langle \psi_{\text{final},i} | \psi_{\text{target},i} \rangle \right|^2 \qquad \text{[Eq. C1]}$$

- Evolve each of the six cardinal states through the **whole** pulse.
- Compare it with where the ideal gate sends it. Only $t = T$ matters.
- The cost is averaged over all six cardinal states, so it also checks the relative phase between $|0_L\rangle$ and $|1_L\rangle$, not just the populations.
- This is **all** an ordinary ("Ord") gate is optimised for.

In plain words: C_1 evaluates where the logical states end up at the end of the gate, not how they got there. A photon lost halfway through is invisible to it.

C_2: transparency at every step

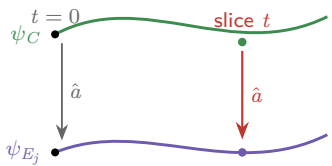
$$F_{\mathrm{ET}} = \frac{1}{N}\sum_{i=0}^{N-1} \frac{\left|\langle\psi_{E_j}(t_i)|\hat{a}|\psi_C(t_i)\rangle\right|^2}{\mathcal{N}}, \qquad \mathcal{N} = \sqrt{\langle\psi_C(t_i)|\hat{a}^\dagger\hat{a}|\psi_C(t_i)\rangle} \qquad \text{[Eq. C2]}$$

Average over the six cardinal pairs: $C_2 = 1 - F_{\mathrm{ET,avg}}$. [Eq. C3]

Evolve ψ_C and ψ_{E_j} **separately**, both from $t = 0$. At each step:

- $\hat{a}\,|\psi_C(t)\rangle$: where a loss *now* would send you.
- $|\psi_{E_j}(t)\rangle$: where a loss at $t = 0$ has been carried by now.

Overlap $= 1$ means the two agree.



We divide by $\mathcal{N}^2$, not $\mathcal{N}$, so that $F_{\mathrm{ET}} \leq 1$ (Cauchy–Schwarz). See notes.

C_3: keep the dynamics spread out

The loophole. A state that barely moves is nearly transparent for free. So the optimiser can idle for most of the pulse and do the whole gate in a short burst at the end. C_2 averages over every step, so it reads high, although the burst is not transparent.

The fix. Measure the speed of the state through Hilbert space, and penalise *uneven* speed:

$$v_i = \frac{2}{\Delta t} \arccos \left| \langle \psi_C(t_i) | \psi_C(t_{i+1}) \rangle \right| \qquad \text{[Eqs. C4–C5]}$$

$$\sigma_j^2 = \frac{1}{N} \sum_{i=0}^{N-1} \left(v_i - \bar{v} \right)^2, \qquad C_3 = \frac{1}{6} \sum_{j=1}^{6} \sigma_j^2 \qquad \text{[Eqs. C6–C7]}$$

- $2 \arccos |\langle \cdot | \cdot \rangle|$ is the distance between neighbouring states, so v is a speed.
- A **regulariser**: it does not improve transparency by itself.

We use $\sigma_j^2 / \bar{v}^2$, which has no units. See notes.

Putting it together: two stages

Stage	(w_1, w_2, w_3)	Goal	What happens
1	$(1,\ 0.6\text{–}0.75,\ 5\text{–}10)$	build transparency	F_{ET} up to 0.83, fidelity 0.9–0.99
2	$(1,\ 0.1,\ 0)$	finish the gate	transparency changes by $\sim 1\%$

[App. C]

The control experiment

Ord = the same optimiser with C_1 alone. EsT and Ord differ only in the cost, so a transparency gap can be pinned on the cost, but only once it is larger than the spread between random starts.

Hardware constraints on the pulse

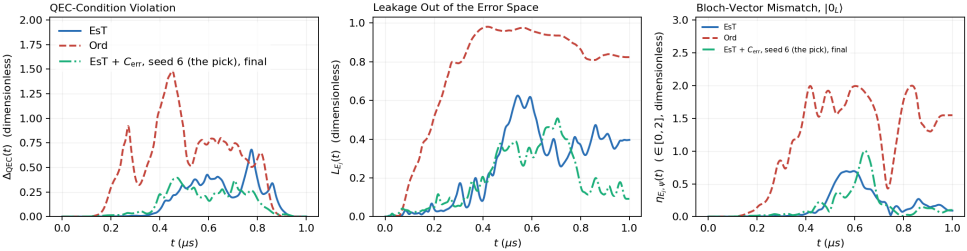
Constraint	Value	Purpose
Bandwidth	± 50 MHz about the carrier	matches the AWG; Fourier-cut the pulse
Amplitude	$\lvert\epsilon\rvert/2\pi,\ \lvert\Omega\rvert/2\pi \leq 4$ MHz	maximum drive strength
Ramp	48 ns Gaussian rise and fall	start and end at zero drive
T gate	cavity drive off	transmon only

[App. C]

How they are imposed

In our code the shape constraints are built into the pulse: $x \to$ band-limit $\to$ ramp $\to$ mask $\to u$. The amplitude cap is a *circle* in the I–Q plane (slide 34), because the amplitude of the drive is the magnitude of the vector (I, Q), i.e. $\sqrt{I^2 + Q^2}$. [ours]

Adding an error-space term: the end-of-gate leak

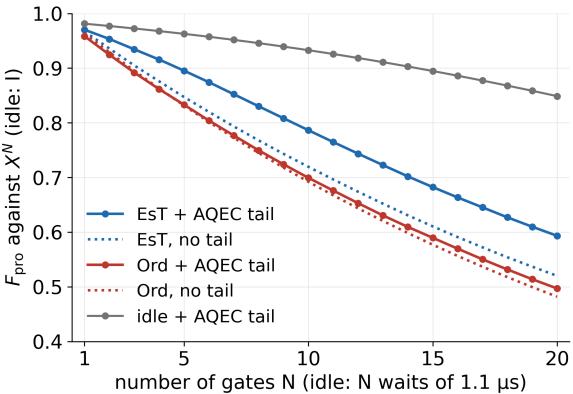


Δ_{QEC}, L as on slide 25, η for $|0_L\rangle$ only; same EsT (blue) and Ord (red) pulses. Green: EsT + C_{err} (error-space fidelity at $t = T$, weight 0.3, stage 1 only), the gate a pick rule chose from 13 seeds: seed 6, final two-stage gate (`est_optimization.ipynb` Fig. 7c). [ours]

X gate	EsT, seed 0	Ord, seed 0	EsT + C_{err}, seed 6 (pick)
$L(T)$, end of gate	0.396	0.824	**0.092**
Δ_{QEC}, peak	0.68	1.49	**0.40**
F_1	0.9993	1.0000	0.9996

Caveats: the pick is 1 of 13 seeds (3 parked, only seed 6 passed the rule); Fock levels 9 / 8 / 9; paper values digitised or read by eye.

Repeated X gates, then one AQEC tail



- N X gates, then **one** 0.8 μs AQEC tail, trained once after one gate.
- **EsT gains more with N:** 0.59 vs 0.52 at $N = 20$. A loss during an EsT gate can still be repaired.
- **Ord gains almost nothing** (0.50 vs 0.48).
- Waiting with the same tail is still best (0.85).

Ours (`FWM_tails.ipynb` Fig. 8a): Shirol et al.'s chip and code, our EsT and Ord X gates (1.1 μs), open system. AQEC tail: two trained pump combs, not the paper's AQEC pulse. Idle: Kerr-corrected decoding.

Our replication: status and open questions

Done (simulation only)

- Independent model and cost, with two gradient codes (autodiff and hand-derived) that agree.
- X, H, T trained, EsT and Ord, with two documented changes to the printed cost (C_2, C_3).
- On X and H, EsT lowers leakage and trajectory mismatch about 3–7$\times$ against Ord (one seed; slide 26).
- Extra costs tried ($\Delta\bar{n}$, end-of-gate error-space fidelity, smoothness): each moves its own target; none is a free win.

Open

- Seed-to-seed spread of the EsT–Ord gap.
- End-of-gate leakage: $L(T) \approx 0.4$ on X, against the paper's 0.013. An error-space term gets 0.09, at 1 seed of 13 (slide 43).
- Transmon errors are outside the model.
- A Fock-matched, multi-seed comparison with the paper.

Setup: code space, projector, errors

A logical qubit lives in a **code space** $\mathcal{C} \subset \mathcal{H}$, with orthonormal codewords $\{|c_i\rangle\}$ and projector

$$P_{\mathcal{C}} = \sum_i |c_i\rangle\langle c_i| = UU^\dagger, \qquad P_{\mathcal{C}}^\dagger = P_{\mathcal{C}}, \quad P_{\mathcal{C}}^2 = P_{\mathcal{C}},$$

where U is the encoding isometry ($U^\dagger U = \mathbb{I}_{\text{logical}}$).

Noise is a channel in **Kraus form**:

$$\varepsilon(\rho) = \sum_j E_j \rho E_j^\dagger, \qquad \sum_j E_j^\dagger E_j = \mathbb{I}.$$

The Kraus set is not unique

Any unitary recombination $F_k = \sum_j W_{kj} E_j$ gives the *same* channel. We will use this freedom.

The Knill–Laflamme theorem

Statement

A recovery operation correcting $\{E_a\}$ exists **if and only if** there is a Hermitian matrix $\alpha = (\alpha_{ab})$, $\alpha_{ab} \in \mathbb{C}$, with

$$P_{\mathcal{C}}\, E_a^\dagger E_b\, P_{\mathcal{C}} = \alpha_{ab}\, P_{\mathcal{C}} \qquad \forall a, b.$$

Equivalently, in codeword form:

$$\left\langle i_L \middle| E_a^\dagger E_b \middle| j_L \right\rangle = \alpha_{ab}\, \delta_{ij}.$$

Two independent demands hide in that one equation:

- δ_{ij}: errors must not **mix** codewords with each other.
- α_{ab} *independent of* i: the environment must learn nothing about **which** codeword it hit.

Why that condition is enough (sketch)

1. Diagonalise α and recombine the errors to match (same channel). Then

$$P_{\mathcal{C}}\, F_k^\dagger F_l\, P_{\mathcal{C}} = d_k\, \delta_{kl}\, P_{\mathcal{C}}.$$

2. So each error moves the code **rigidly** onto its own copy $\mathcal{C}_k$, and different copies are **orthogonal**.
3. **Recovery:** measure which copy (the syndrome), then rotate it back. Outcome k has probability d_k for every logical state.

In plain words: errors scatter the code into orthogonal, undistorted copies of itself. Find the copy, rotate it back. The measurement learns the error, never the logical state.

Full derivation: backup slides B4–B6.

Why that condition is enough (1/2): untangle the errors

Shorthand on these backup slides: $P \equiv P_{\mathcal{C}}$.

1. α is Hermitian, so a unitary u diagonalises it:

$$\alpha = u\, d\, u^{\dagger}, \qquad d = \mathrm{diag}(d_k), \quad d_k \geq 0.$$

2. Recombine the Kraus operators with it (same channel, slide B1): $F_k = \sum_a u_{ak}\, E_a$.

3. The error-correction condition becomes

$$P\, F_k^{\dagger} F_l\, P = (u^{\dagger} \alpha\, u)_{kl}\, P = d_k\, \delta_{kl}\, P.$$

Only the diagonal survives

Different errors $F_k \neq F_l$ no longer overlap on the code, and error k occurs with probability d_k whichever codeword it hits.

Why that condition is enough (2/2): undistorted copies

Normalise each error on the code:

$$V_k \equiv \frac{1}{\sqrt{d_k}} F_k P.$$

Each V_k is a **partial isometry**: it maps $\mathcal{C}$ onto an image $\mathcal{C}_k = V_k \mathcal{C}$ without distorting it, and

$$V_k^\dagger V_k = P, \qquad V_k^\dagger V_l = 0 \;\; (k \neq l).$$

So the errors scatter the code into **disjoint, orthogonal, undistorted copies** of itself.

Correcting a discrete set also corrects its entire linear span, so discrete errors suffice.

Recovery: find the copy, rotate it back

After the noise, the code sits on copy k with probability d_k: $\varepsilon(\rho) = \sum_k d_k \, V_k \, \rho \, V_k^\dagger$.

1. **Error-space projector** onto $\mathcal{C}_k = V_k \mathcal{C}$:

$$P_k = V_k \, V_k^\dagger, \qquad P_k P_l = \delta_{kl} \, P_k.$$

2. **Measure** $\{P_k\}$. Outcome k and the state it leaves:

$$p_k = \mathrm{Tr}\big[P_k \, \varepsilon(\rho)\big] = d_k, \qquad \varepsilon(\rho) \;\to\; V_k \, \rho \, V_k^\dagger.$$

3. **Rotate back** with $V_k^\dagger$:

$$V_k^\dagger \big(V_k \, \rho \, V_k^\dagger \big) \, V_k = P \rho P = \rho.$$

$p_k = d_k$ for every ρ: the syndrome reveals the error, never the logical state.

The binomial “kitten” code

$$|0_L\rangle = \frac{|0\rangle + |4\rangle}{\sqrt{2}}, \qquad |1_L\rangle = |2\rangle$$

Both codewords live on the **even-parity** Fock states.

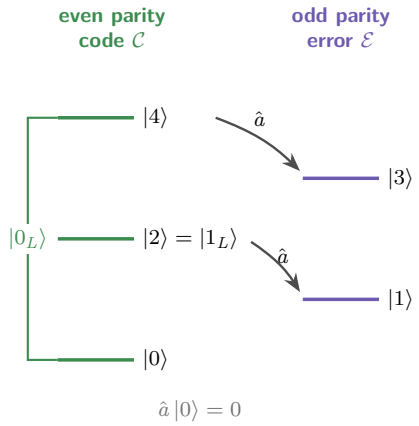
A single photon loss $\hat{a}$ sends them to

$$\hat{a}\,|0_L\rangle = \sqrt{2}\,|3\rangle \equiv \sqrt{2}\,|0_E\rangle\,,$$

$$\hat{a}\,|1_L\rangle = \sqrt{2}\,|1\rangle \equiv \sqrt{2}\,|1_E\rangle\,,$$

entirely on the **odd-parity** states.

[Sec. III]



Why the kitten code satisfies Knill–Laflamme

Two facts, and they are exactly the two demands from the theorem:

1. **Distinct, orthogonal images.** $|0_L\rangle \to |0_E\rangle$, $|1_L\rangle \to |1_E\rangle$ with $\langle 0_E|1_E\rangle = \langle 3|1\rangle = 0$. Loss acts as an **isometry** onto the error space, so codewords do not mix.
2. **Equal coefficients.** Both images carry the same $\sqrt{2}$, i.e.

$$\left\langle 0_L \middle| \hat{a}^\dagger \hat{a} \middle| 0_L \right\rangle = \left\langle 1_L \middle| \hat{a}^\dagger \hat{a} \middle| 1_L \right\rangle = 2.$$

   Equal mean photon number $\Rightarrow$ no which-codeword information.

Together:

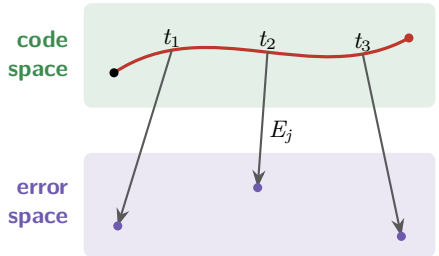
$$P_{\mathcal{C}}\, \hat{a}^\dagger \hat{a}\, P_{\mathcal{C}} = 2\, P_{\mathcal{C}} \qquad \text{(proportional to the logical identity).}$$

In plain words: the environment learns *that* a photon left, and nothing about *which* state it left from. No which-path information, so no logical dephasing (to first order in κt).

Why a gate breaks all of that

Idling. The codewords barely move (only small Kerr phases). A lost photon lands on nearly the same known error word, parity detects it, and you recover.

During a gate. The drive moves the state, so mid-gate it is no longer a codeword.



Photon lost at t: $\underbrace{U(T,t)}_{\text{finish}}\underbrace{E_j}_{\text{loss}}\underbrace{U(t,0)}_{\text{start}}|\psi_L(0)\rangle$ depends on t, and t is **random**.

In plain words: a photon lost mid-gate ends up somewhere different for every loss time. The recovery at the end has one fixed target, so it cannot undo them all.

When does a loss commute through? Error closure

Push the jump past the Hamiltonian. A loss at any time stays correctable if, at every instant, the commutator lands back among the correctable errors:

$$\big[\, H(t),\, \hat{a} \,\big] \;\in\; \mathrm{span}\{\text{correctable errors}\} \qquad [\text{App. A}]$$

This is *error closure*. Exact transparency, $[H, \hat{a}] = \lambda\, \hat{a}$, is the special case.

Term by term, for the Hamiltonian of slide 14:

Term	**Commutator with $\hat{a}$**	**Correctable?**
$\omega_a\, \hat{a}^\dagger \hat{a}$, $\chi\, \hat{a}^\dagger \hat{a}\, \hat{q}^\dagger \hat{q}$	$\propto \hat{a}$ (times $\hat{q}^\dagger \hat{q}$)	yes
linear drives $\epsilon\, \hat{a} + \Omega\, \hat{q} + \mathrm{h.c.}$	$-\epsilon^*$, a number	yes
Kerr K, higher dispersive χ'	$\propto \hat{a}^\dagger \hat{a}^2 = \hat{n}\, \hat{a}$	no

The obstruction: nonlinear terms that will not commute

The system Hamiltonian (slide 14) carries the cavity self-Kerr $\frac{K}{2}\hat{a}^{\dagger 2}\hat{a}^2$ and a higher-order dispersive term $\frac{\chi'}{2}\hat{a}^{\dagger 2}\hat{a}^2\hat{q}^\dagger\hat{q}$. Their commutator with the jump operator is

$$\left[\frac{K}{2}\hat{a}^{\dagger 2}\hat{a}^2 + \frac{\chi'}{2}\hat{a}^{\dagger 2}\hat{a}^2\,\hat{q}^\dagger\hat{q},\ \ \hat{a}\right] = -\left(K + \chi'\,\hat{q}^\dagger\hat{q}\right)\hat{a}^\dagger\hat{a}^2. \qquad \text{[Eq. A6]}$$

- $\hat{a}^\dagger\hat{a}^2 = \hat{n}\,\hat{a}$ is a jump weighted by photon number. It hits $|0_L\rangle$ and $|1_L\rangle$ with different strengths (3 vs 1), so it violates the "α independent of i" half of Knill–Laflamme.
- A linear drive can soften its effect along a trajectory, but cannot cancel it: its own commutator with $\hat{a}$ is just a number, with no $\hat{n}$-dependent part (App. A).

Therefore: error *semi*transparent

Exact ET is not expected with linear drives alone. Accept a small, nonzero violation in exchange for simple, standard control, and then **measure how much you recover**.

Letting the code space move

The conditions for *static* ET ask the state to stay inside the fixed $\mathcal{C}(0)$ and $\mathcal{E}(0)$. Gate drives sweep it far outside both. So: **track moving subspaces** instead.

Instantaneous code space [Eq. 1]

$$\mathcal{C}(t) = \left\{ U(t,0)\,|\psi_L(0)\rangle \;:\; |\psi_L(0)\rangle \in \mathcal{C}(0) \right\}, \qquad P_{\mathcal{C}}(t) = U(t,0)\,P_{\mathcal{C}}\,U^{\dagger}(t,0).$$

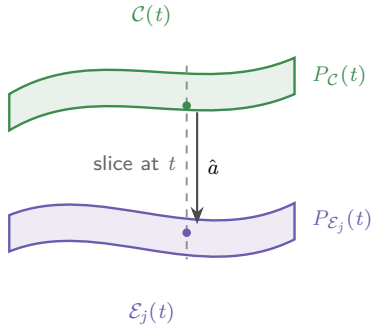
- A 2-dimensional subspace carried along the **ideal trajectory**.
- $U(t,0)$ is the *ideal, lossless* propagator of the driven system.

And the error space with it

Instantaneous error space [Eq. 2]

$$\mathcal{E}_j(t) = \mathrm{span}\Big\{\, E_j\,|\psi_L(t)\rangle \;:\; |\psi_L(t)\rangle \in \mathcal{C}(t)\,\Big\}$$

$$P_{\mathcal{E}_j}(t) \;\propto\; E_j\, P_{\mathcal{C}}(t)\, E_j^\dagger$$



For photon loss, $E_j = \hat{a}$:

$$\mathcal{E}(t) = \mathrm{span}\Big\{\hat{a}\,U(t,0)\,|0_L\rangle\,,\;\hat{a}\,U(t,0)\,|1_L\rangle\Big\}$$

In plain words: where a photon loss *right now* would send you, given the ideal trajectory.

The three conditions

The dynamic commuting square holds if **three things are true at every instant**:

1. Correctable

The moving code space is still a valid QEC code.

Knill–Laflamme, instant by instant

2. Tracking

The error space goes where the drive carries it.

evolved space = instantaneous space

3. Matched

Code and error spaces feel the same internal dynamics.

same generator, up to a phase rate

Each one gets an equation, and later a number that measures how badly it is violated.

Condition 1: instantaneous correctability

Over the error set $E = \{\mathbb{I},\ \hat{a}\}$, every projected error product must be **proportional to the logical identity**:

$$P_{\mathcal{C}}(t)\, E_i^{\dagger} E_j\, P_{\mathcal{C}}(t) = c_{ij}(t)\, P_{\mathcal{C}}(t), \qquad \forall i, j, t. \qquad \text{[Eq. 3]}$$

This is Knill–Laflamme again, now demanded of the code space *as it moves*.

The kitten code at $t = 0$:

$$c(0) = \begin{pmatrix} c_{II} & c_{Ia} \\ c_{aI} & c_{aa} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 2 \end{pmatrix}$$

At $t = 0$ the only live entry is c_{aa}, and $\hat{a}^{\dagger}\hat{a} = \hat{n}$, so it is the demand

$$\langle 0_L|\hat{n}|0_L\rangle = \langle 1_L|\hat{n}|1_L\rangle \ \ (= 2).$$

Condition 2: subspace tracking

$$U(t,0)\,\mathcal{E}_j(0) = \mathcal{E}_j(t), \qquad \forall j, t. \qquad \text{[Eq. 4]}$$

Two different objects that must coincide:

- **Evolved** $U(t,0)\mathcal{E}_j(0)$: where a jump at $t = 0$ has been *carried to* by now.
- **Instantaneous** $\mathcal{E}_j(t)$: where a jump *right now* would land.

Why anchoring at $t = 0$ is enough

A jump can happen at any time s. Right after it the state is in $\mathcal{E}_j(s)$, and the drive carries it to $U(T,s)\,\mathcal{E}_j(s)$ by the end T.

$$U(T,s)\,\mathcal{E}_j(s) \underset{\text{Eq. 4 at } s}{=} U(T,s)\;U(s,0)\,\mathcal{E}_j(0) \underset{\text{compose}}{=} U(T,0)\,\mathcal{E}_j(0) \underset{\text{Eq. 4 at } T}{=} \mathcal{E}_j(T).$$

So an error j always ends in the same subspace $\mathcal{E}_j(T)$, whenever it happened.

Condition 3: matched internal dynamics

With $P_j(t) \propto E_j P_{\mathcal{C}}(t)$ the map carrying the code space onto the instantaneous error space (an isometry up to normalisation),

$$P_j^\dagger(t)\, H(t)\, P_j(t) = P_{\mathcal{C}}^\dagger(t)\, H(t)\, P_{\mathcal{C}}(t) + d_j(t)\, P_{\mathcal{C}}(t), \qquad \forall j, t, \qquad \text{[Eq. 5]}$$

where $d_j(t)$ is a number. The paper calls it complex; both sides are Hermitian, so it is real.

Read as Hamiltonians restricted to the two subspaces:

$$H_{\mathcal{E}}(t) = H_{\mathcal{C}}(t) + d_j(t)\, \mathbb{I}.$$

- The two subspaces are driven by the **same generator**, differing only by a multiple of the identity.
- A shift proportional to $\mathbb{I}$ is a global phase rate, physically invisible.

In plain words: the drive must push the error space around in exactly the same way it pushes the code space, apart from an overall phase.

Three metrics, one per condition

Each condition gets a scalar that is **zero when the condition holds**.

Condition	Metric	Question it answers
1	QEC violation $\Delta_{\mathrm{QEC}}(t)$ (Eq. 6)	Is the moving code still correctable?
2	Leakage $L_{\mathcal{E}_j}(t)$ (Eq. 7)	Is the error state in the right subspace?
3	Trajectory mismatch $\eta_{\mathcal{E}_j,\psi}(t)$ (Eq. 8)	Is it in the right place inside it?

All three are **time-dependent**: they are evaluated at every step of the pulse, not just at $t = T$.

Metric 1: QEC violation $\Delta_{\mathrm{QEC}}(t)$

Form the 2×2 projected error matrices, over $E = \{\mathbb{I}, \hat{a}\}$:

$$M_{ik}(t) = P_{\mathcal{C}}(t)\, E_i^\dagger E_k\, P_{\mathcal{C}}(t) = c_{ik} P_{\mathcal{C}} + x_{ik} X_{\mathcal{C}} + y_{ik}\, Y_{\mathcal{C}} + z_{ik} Z_{\mathcal{C}}. \qquad \text{[Eqs. A8–A9]}$$

The $P_{\mathcal{C}}$ part is correctable; the X, Y, Z parts are **not**. So strip the identity and measure what is left:

$$\Delta_{\mathrm{QEC}}(t) = \sum_{i,k} \left\| M_{ik} - \tfrac{1}{2} \operatorname{Tr}[M_{ik}]\, P_{\mathcal{C}} \right\|_2 = \sum_{i,k} 2\big(|x_{ik}|^2 + |y_{ik}|^2 + |z_{ik}|^2\big). \qquad \text{[Eqs. 6, A11]}$$

In plain words: $\Delta_{\mathrm{QEC}}(t) = 0$ means a photon lost at time t can still be corrected. Larger means a bigger uncorrectable part. It is a size, not a probability.

$\|A\|_2 = \operatorname{Tr}(A^\dagger A)$ is the squared Frobenius norm (App. A): no square root, hence the factor of 2.

Metric 2: instantaneous leakage $L_{\mathcal{E}_j}(t)$

$$L_{\mathcal{E}_j}(t) = 1 - \mathrm{Tr}\big[\, P_{\mathcal{E}_j}(t)\, \rho(t) \,\big] \qquad \text{[Eq. 7]}$$

where $\rho(0)$ is the **maximally mixed state in the static error space** $\mathcal{E}_j(0)$, evolved unitarily with no losses:

$$\rho(0) = \frac{P_{\mathcal{E}_j}(0)}{d} = \tfrac{1}{2}\big(|0_E\rangle\langle 0_E| + |1_E\rangle\langle 1_E|\big), \qquad \rho(t) = U(t,0)\, \rho(0)\, U^\dagger(t,0).$$

So the trace compares two subspaces: the **evolved** error space (carried in $\rho(t)$) against the **instantaneous** one (in $P_{\mathcal{E}_j}(t)$). That is exactly Condition 2.

In plain words: the probability, averaged over all logical inputs, that a state which suffered the error at $t = 0$ is found *outside* the subspace an error at time t would produce.

Metric 3: trajectory mismatch $\eta_{\mathcal{E}_j,\psi}(t)$

$$\eta_{\mathcal{E}_j,\psi}(t) = \left\| \vec{r}_{\mathcal{C}}(t) - \vec{r}_{\mathcal{E}_j}(t) \right\|_2 \in [0,2] \qquad \text{[Eq. 8]}$$

$$\vec{r}_{\mathcal{C}}(t) = \mathrm{Tr}\left[\vec{\sigma}_{\mathcal{C}}(t)\,\rho_{|\psi\rangle}(t)\right], \qquad \vec{r}_{\mathcal{E}_j}(t) = \mathrm{Tr}\left[\vec{\sigma}_{\mathcal{E}_j}(t)\,\tilde{\rho}_{\left|\psi_{E_j}\right\rangle}(t)\right]$$

- Take a logical state $|\psi\rangle$ and its error partner $\left|\psi_{E_j}\right\rangle \propto E_j\,|\psi\rangle$; evolve both.
- Write each as a Bloch vector in **its own** instantaneous logical frame ($\vec{\sigma}_{\mathcal{C}}(t)$ in the code space, $\vec{\sigma}_{\mathcal{E}_j} \propto E_j \vec{\sigma}_{\mathcal{C}} E_j^\dagger$ in the error space), and measure the distance.
- The error state is *projected into* $\mathcal{E}_j(t)$ first, so $\eta_{\mathcal{E}_j,\psi}$ is blind to leakage: independent information from $L_{\mathcal{E}_j}$.
- Evaluated over all 6 Bloch cardinal states.

In plain words: "where is the state on the code sphere?" vs. "where is it on the error sphere?" Zero means no relative rotation (Condition 3).

Reading the Hamiltonian

- **Dispersive χ.** $(\omega_q + \chi\, \hat{a}^\dagger \hat{a})\, \hat{q}^\dagger \hat{q}$: the transmon frequency moves by χ per photon, so a transmon drive is *photon-number selective*.
- **Kerr K, K', χ'.** Small photon-number-dependent shifts. K and χ' (and, weakly, K') are what break error transparency (Eq. A6).
- **Rotating frame.** Removes ω_a, ω_q; the static part left over (χ, Kerr, K_q) is the H_0 our simulation uses.
- **Controls.** Only the complex $\epsilon(t)$ and $\Omega(t)$, which GRAPE shapes.
- **The error the code targets.** Cavity photon loss: jump operator $\hat{a}$, rate $\kappa = 1/T_1$.

Notation clash

$\epsilon(t)$ is a *drive amplitude*. The noise *channel* of the error-correction slides is written $\varepsilon(\rho)$, a different object.